Autoimmunity and Related Diseases

Loss of self-tolerance, mechanisms, and management

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By the end of this lecture you should be able to

1. Define autoimmunity and explain how it differs from autoimmune disease
2. Describe central and peripheral mechanisms of self-tolerance
3. Explain how immune privilege protects vital organs
4. Outline genetic and environmental triggers of autoimmunity
5. Compare antibody, T-cell and immune complex mechanisms
6. Recognise key autoantibodies and their disease associations
7. Outline the principles of immunosuppressive management

Autoimmunity versus autoimmune disease

Immune responses to self occur normally; disease is the consequence of harm

Autoimmunity

An immune response directed against self-antigens, also called autoantigens.

Low-level autoreactivity is found in many healthy people, so it is not automatically disease.

Autoimmune disease

Tissue injury or dysfunction caused by a loss of self-tolerance, through autoantibodies or autoreactive T cells.

Diagnosis needs both evidence of an autoimmune process and organ damage or dysfunction.

Central tolerance: testing in the thymus

Developing lymphocytes are tested against self before they are released

1

Develop

Immature T cells in the thymus, and B cells in the bone marrow, express random receptors

2

Test

Thymic epithelium presents self-antigens; the AIRE gene helps express many tissue-specific proteins there

3

Delete

Cells that bind self strongly undergo negative selection (apoptosis)

4

Edit or release

B cells can change their receptor (receptor editing); escapees are controlled by peripheral mechanisms

Peripheral tolerance: controlling escapees in tissues

1	**Anergy**	A T cell that sees antigen without co-stimulation becomes functionally inactive rather than activated.
2	**Regulatory T cells**	Tregs ($CD4^+$ $CD25^+$ $FOXP3^+$) suppress effector cells using IL-10, TGF-β and CTLA-4.
3	**Clonal deletion**	Persistent autoreactive activation drives apoptosis, removing the reactive clone.

Clinical link: a loss of Treg function is one of the most consistently implicated defects in autoimmune disease.

Immune privilege: sites that limit immune responses

Antigens here can be recognised as foreign, yet the response is restrained

Site	Why it is protected	Risk if exposed
Brain and spinal cord	Blood–brain barrier limits immune cells and antibodies	Neuronal antigens released after injury
Eye	Blood–ocular barrier; immune damage would destroy retina and cornea	Sympathetic ophthalmia
Testes	Blood–testis barrier hides sperm antigens	Autoimmune orchitis and infertility
Uterus and fetus	Placenta and local Tregs protect the semi-allogeneic fetus	Pregnancy complications

How immune privilege is maintained

Physical barriers

Tight endothelial junctions and the blood–brain, blood–retina and blood–testis barriers restrict entry of immune cells and antibodies.

Lymphatic absence

Without lymphatic drainage, antigens are not carried to lymph nodes, so they are poorly presented to naive T cells.

Local suppression

TGF-β and IL-10 dampen activation, while Fas ligand (FasL) triggers apoptosis of infiltrating Fas-bearing lymphocytes.

Local Tregs

Regulatory T cells maintain tolerance and limit local inflammation.

When immune privilege breaks down

Hidden antigens become visible to a primed immune system

Sympathetic ophthalmia	Penetrating eye injury releases sequestered ocular antigens. The immune system attacks the injured eye and can then attack the fellow eye.
Autoimmune orchitis	Testicular trauma or infection exposes sperm antigens, causing inflammation and damage to germ cells, with infertility.
Post-CNS injury inflammation	Damage to the blood–brain barrier exposes neuronal antigens to circulating immune cells.

Principle: privilege delays tolerance failure; it does not guarantee it.

Genetic factors: HLA is the strongest susceptibility locus

HLA class II molecules present peptides to $CD4^+$ T cells; allele differences change which peptides are displayed

Allele	Disease association	Interpretation
HLA-B27	Ankylosing spondylitis	Strongest single association for spondyloarthritis; most carriers never develop it
HLA-DR3, DR4	Type 1 diabetes, rheumatoid arthritis	DR4 is linked to RA; DR3 and DR4 together raise T1D risk
HLA-DR2	SLE, multiple sclerosis	DR2 is associated with both diseases

Genes give risk, not destiny: penetrance is incomplete, and environment and chance shape outcome.

Environmental triggers: infection and drugs

1

Infection

- Molecular mimicry: a microbial antigen resembles a self-antigen (e.g. streptococcal M protein and cardiac myosin in rheumatic fever)
- Bystander activation: inflammation activates autoreactive cells nearby

2

Drugs

- Hydralazine and procainamide can cause a lupus-like syndrome
- Drug-induced disease often resolves when the drug is stopped

3

Other factors

- UV light can release nuclear antigens in skin (SLE)
- Smoking is linked to RA and its antibody-positive form
- Stress and hormones influence disease activity, and women are more often affected

Immunological mechanisms that sustain autoimmunity

1	**Loss of regulatory control**	Reduced Treg number or function allows effector clones to escape suppression.
2	**Abnormal antigen presentation**	Over-expression of MHC or co-stimulatory molecules on tissue cells can activate T cells that would otherwise ignore them.
3	**Epitope spreading**	Tissue damage releases new self-antigens. The immune response then broadens to these, so disease can progress even after the first trigger is gone.

Epitope spreading explains why some autoimmune diseases progress over time without a continuing trigger.

Immunopathogenesis: four mechanisms of injury

Destruction, complex deposition, cell killing or receptor stimulation

Type II · Cytotoxic

Antibody binds a cell surface antigen; complement or phagocytosis destroys the cell

Example: Autoimmune haemolytic anaemia

Type III · Immune complex

Soluble antigen–antibody complexes deposit in vessels and tissues and activate complement

Example: SLE

Type IV · Cell-mediated

Effector T cells cause inflammation and direct cytotoxicity over days

Example: Type 1 diabetes, MS

Type V · Receptor stimulation

Antibody binds a receptor and activates it without killing the cell

Example: Graves' disease

Antibody-mediated disease

Destruction or receptor blockade, depending on the antibody

1

Haemolytic anaemia

- Autoantibodies coat red cells
- Complement and splenic macrophages remove them
- Haemolysis, anaemia and a positive direct Coombs test

2

Myasthenia gravis

- Antibodies target the acetylcholine receptor at the neuromuscular junction
- Receptors are blocked and internalised, reducing the signal
- Fatigable weakness that worsens with activity

3

Graves' disease

- Antibodies bind and stimulate the TSH receptor
- Thyroid hormone production rises, independent of pituitary TSH
- Hyperthyroidism with a diffusely enlarged gland

T-cell mediated disease: inflammation and cell killing

$CD4^+$ helper T cells	Recognise self-peptide on MHC class II and release cytokines such as IFN-γ and IL-17 that recruit and activate other cells.
$CD8^+$ cytotoxic T cells	Recognise self-peptide on MHC class I and kill target cells directly, for example pancreatic β-cells.
Examples	Type 1 diabetes: insulitis destroys β-cells over time. Hashimoto's: lymphocytic infiltration destroys thyrocytes. Multiple sclerosis: demyelination in the CNS.

Hashimoto's is mainly T-cell driven, but antibodies to TPO and thyroglobulin are useful markers of the same process.

Immune complex disease: SLE as the model

1

Clearance fails

Apoptotic nuclear material is not cleared efficiently, so nuclear antigens persist

2

Complexes form

Anti-dsDNA and other antibodies bind these antigens in the circulation

3

Deposition

Complexes lodge in glomeruli, skin and synovium

4

Inflammation

Complement activation and Fc receptor signalling drive injury, with low C3 and C4 as a clue

Clinical picture: rash, arthritis, nephritis and cytopenias reflect where complexes deposit and what they consume.

Classification: organ-specific versus systemic

The pattern predicts treatment choice and monitoring

Feature	Organ-specific	Systemic
Distribution	Confined to one organ or tissue	Widespread, many organs
Typical antigen	Organ-restricted (thyroglobulin, β-cell antigens, AChR)	Ubiquitous nuclear or joint antigens
Examples	Hashimoto's, Graves', type 1 diabetes, myasthenia gravis	SLE, rheumatoid arthritis, systemic sclerosis, Sjögren's
Treatment focus	Replace lost function, or control the organ-specific response	Immunosuppression with multi-organ monitoring

Some patients have overlapping features, so the pattern is a guide rather than a strict rule.

Major autoimmune diseases at a glance

Disease	Target or mechanism	Main features
SLE	Immune complex deposition (Type III)	Rash, arthritis, nephritis, cytopenias
Rheumatoid arthritis	Synovial inflammation (T cells, macrophages)	Symmetrical arthritis, joint deformity
Hashimoto's thyroiditis	T cells and antibodies destroy thyrocytes	Hypothyroidism, goitre
Graves' disease	Anti-TSH receptor antibodies stimulate (Type V)	Hyperthyroidism
Type 1 diabetes	T-cell destruction of β-cells	Insulin deficiency
Myasthenia gravis	Anti-AChR antibodies block the NMJ	Fatigable muscle weakness
Multiple sclerosis	T-cell predominant demyelination	Neurological deficits

Laboratory investigation: autoantibodies in context

Serology supports a diagnosis; it does not make one on its own

Test	Target	Interpretation
ANA	Antinuclear antibody	Sensitive but not specific. Positive in SLE and many healthy people
Anti-dsDNA	Double-stranded DNA	Specific for SLE; titres can track disease activity
Anti-CCP	Cyclic citrullinated peptide	Specific for rheumatoid arthritis; often present before symptoms
Anti-TPO / Tg	Thyroid antibodies	Confirm thyroid autoimmunity in Hashimoto's; TSH receptor antibodies in Graves'

Other tests: low complement in immune complex disease; raised ESR and CRP; organ function (TSH, glucose, renal function).

Management: principles and drug classes

Principles

- There is no universal cure
- Control the immune response and limit organ damage
- Remove triggers such as drugs or infection
- Replace lost function where possible

Immunosuppression

- Corticosteroids reduce activation quickly
- Methotrexate, azathioprine and cyclophosphamide inhibit lymphocyte proliferation
- Monitor for infection and marrow toxicity

Targeted therapy

- Anti-TNF (adalimumab) blocks a key cytokine
- Rituximab depletes $CD20^{+}$ B cells
- Anti-IL-6 blocks signalling in inflammatory disease
- Plasmapheresis or IVIG remove or neutralise antibodies (short term)

Prognosis and key take-home messages

Prognosis

- Depends on disease, organ involvement and treatment response
- Early diagnosis and tailored treatment improve outcomes
- Long-term follow-up for relapse and infection risk
- Screen for secondary autoimmune disease

Take-home messages

- Autoimmunity arises from genetic and environmental factors that break self-tolerance
- Mechanisms are antibody, T cell and immune complex; often they overlap
- Immune privilege protects brain, eye and testis
- Diagnosis combines clinical suspicion with specific serology